Woojin Bae

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EDUCATION

Seoul National University (SNU), Seoul, South Korea *Aug. 2028 (expected)*
M.S. Chemical and Biological Engineering (*CBE*) (PI: *Prof. Yousung Jung)*
Seoul National University, Seoul, South Korea *Aug. 2026*
B.S., *CBE, Cum Laude*
North Carolina State University (NCSU), Raleigh, NC *May 2025*
Exchange student in Chemical and Biomolecular Engineering
Daejeon Science High School for the Gifted, Daejeon, South Korea *Feb. 2020*

PUBLICATIONS & PRESENTATIONS (*†: equally contributed, *: corresponding author*)

W. Bae, H. Jang, Jaewon Bae*, C. Park *, Combining Physical and Retrieved Empirical Priors for Solid-State Synthesis Condition Prediction. *NeurIPS 2026 Workshop AI4Mat.*

K. Jang, D. Kim, **W. Bae**, Jaewon Bae*, C. Park *, A Structure-to-Density Model for Trajectory-Free Lithium Transport Screening. *NeurIPS 2026 Workshop AI4Mat.*

J. Hong†, H. Jang†, **W. Bae**†, Jaewon Bae*, C. Park *, Retro-Forge: A Multi-Step Pairwise Retrosynthesis Framework for Solid-State Materials Synthesis. *ICML 2026 Workshop AI4Science.*

Jinge Xu, Junbin Li, Pragyan Jha, Nikolai Mukhin, **Woojin Bae**, Christopher H.J. Moran, Richard B. Canty, Milad Abolhasani*, Self-Driving Chemistry for Route-Encoded Colloidal Nanoscience. *Submitted.*

W. Bae†, S.Jung†, S. Ji†, Y. Jeon* & J. Park*, SEI Structures Dependent Li-Morphology Characterized by Cryo-TEM. [Poster Presentation]

- *2025 KIChE Fall meeting and International Symposium*, Oct. 15-18, 2025.
- *The 2024 Annual Fall Conference of Korean Society of Microscopy*, Nov. 25-26, 2024.

RESEARCH EXPERIENCE

NanoForge AI - Researcher *Mar. 2026 – Aug. 2026*

Topic: Solid state reaction process prediction leveraging Large Language Model.

- Trained an LLM model to predict precursor selection as a stepwise solid-state reaction planning task.
- Leveraged chemical prior and *Retrieval Augmented Generation* (RAG) to predict optimal calcination and sintering temperatures.

Multi-dimensional Materials Chemistry Lab, SNU *Mar. 2024 – Mar. 2026*
Undergraduate Researcher - (PI: *Prof. Jungwon Park*) *Seoul, Korea*

Topic: SEI Structures Dependent Li-Morphology Characterized by cryo-TEM

- Team Leader; Developed a DBSCAN-based protocol for quantitative image analysis protocol for cryogenic TEM characterization of SEI structures.
- Observed correlation between SEI morphology, crystallinity, and composition; proposed a formation mechanism of Li-deposition. (Co-Advised by *Prof. Won Bo Lee*)

Topic: Computational Modeling of AI-Driven Synthesized Perovskite Nanocrystals

- Awarded a $2,000 research grant by SNU College for a project across SNU and NCSU.

- Initiated and led collaborative project across SNU and NCSU to characterize AI-driven synthesized Mn-doped $CsPb(Cl/Br)_3$ QDs, by TEM analysis and DFT calculation.

Self-Driving Fluidic Lab, NCSU *Jan. 2025 – May 2025*
Undergraduate Researcher - (PI: *Prof. Milad Abolhasani*) *Raleigh, NC*

Topic: Autonomous Mn-doped Perovskite Nanocrystals Synthesis by a Self-Driving Lab

- Achieved Photoluminescence Quantum Yield improvement of Mn-doped $CsPb(Cl/Br)_3$ perovskite through genetic algorithm-based synthesis optimization in autonomous lab.
- Presented a poster at Spring Undergraduate Research & Creativity Symposium.

Summer Undergraduate Initiative (SURI), ASU *May 2025 – Jul. 2025*
Undergraduate Researcher - (PI: *Prof. Sandhya Susarla*) *Tempe, AZ*

Topic: Synthesis of Perovskite QDs for Phonon-Polariton Interaction with hBN

- Selected for the SURI fellowship at Arizona State University ($5,000 stipend).
- Synthesized perovskite nanocrystals with controlled morphologies to investigate phonon–polariton coupling with hexagonal boron nitride (hBN).
- Presented a poster at Summer Research Initiative Final Research Showcase.

HONORS & AWARDS

Honorable Mention | Korea Process Simulation Olympiad 2025, KIChE, Korea *Sep. 2025*

- National finalist with process design for high-purity methanol (PI: *Prof. Jong Min Lee*)

Excellence Research Award | 2024 Student-Directed Research Symposium *Jan. 2025*

KorUS Educational Exchange Initiative for Youth in STEM | MOTIE, Korea *Nov. 2024*

- Awarded national $9,000 scholarship as a STEM-specialized exchange student.

Excellence Award | Undergraduate Research Program for Advanced Equipment *Jan. 2024*

Team Section Winner | We Change SNU Contest, SNU *Jul. 2022*

- A spatial-based messenger allowing messages to be accessed only at SNUfestival locations designated by the sender.

Education and Research Foundation Scholarship | College of Engineering, SNU *Spring 2021*

Future Chemical Talent Award | Korean Union of Chemical Science & Technology *Feb. 2020*

- Awarded as one of the top 6 science high school students in Korea.

LEADERSHIP & ACTIVITIES

Undergraduate Member | Korean-American Scientists and Engineers Association *Jan. 2025*

- Delivered a research presentation at Korean-American undergraduate conference.

Tokyo Forum Youth Session | The University of Tokyo, Japan *Fall 2024*

- Representative of Korea, in a session on Aging Society and Low Birth Rate.

Young Engineers Honor Society | National Academy of Engineering of Korea *Nov. 2024*

- Korean engineering honor society under National Academy of Engineering of Korea.

Entrepreneurship Bootcamp | Hong Kong University of Science and Technology *Jul. 2024*

- Led a global team in a mock-up competition and won third prize for developing a worldwide healthcare platform among international participants.

SNU-UMN Culture Exchange Program |University of Minnesota Twin Cities, MN *Mar. 2024*

Winter Data Science Bootcamp: **Computing** | Graduate School of Data Science, SNU *Feb. 2024*

SNU in EU Union | Brussels School of Governance, Belgium *Jul. 2022*

- Study Abroad Program, delivered a presentation titled "The European External Action Service".

Republic of Korea Air Force | 11th Fighter Wing, Daegu, Korea *Aug. 2022 – May 2024*

Marketing team manager | SNUfestival, Student Council, SNU *Jun. 2021 – May 2022*

- Promoted the festival through the online website and secured $2,200 sponsorship from a shared scooter company during the first in-person festival after the COVID-19 pandemic.

Vice president | Chemical and Biological Engineering Interim Leadership, SNU *Winter 2020*

- Developed an online program for incoming freshmen during the COVID-19 pandemic.

TEACHING & MENTORING EXPERIENCES

Peer Tutor | Basic Computing: Exploring Data for the first time, SNU *Spring 2026*

SNU Corps in Indonesia | Universitas Mataram, Lombok, Indonesia *Feb. 2026*

Korean Language Peer Mentor | Language Education Institute, SNU *Fall 2024*

Peer Tutor | Basic Computing: First Adventures in Computing, SNU *Fall 2024, 2025*

Lecturer, TEM Seminar | Multi-dimensional Materials Chemistry Lab, SNU *May 2024*

Student Mentor, SNU Mentoring | SNU Social Responsibility *Apr. 2023 – Aug. 2025*

TECHNICAL SKILLS & LANGUAGES

Instruments: TEM, SEM, XRD, STEM/EDS, UV-Vis, OM, PL, ICP-MS

Programming Language: Python, MATLAB, C

Software: Gatan DigitalMicrograph, VASP, APS, Adobe, Fusion 360

Languages: Korean (Native), English (C1, TOEFL 103)